

JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
Purchase Request No. NB305000-26-00131

1. Identification of the agency and the contracting activity:

U.S. Department of Commerce, National Institute of Standards and Technology (NIST), Office of Acquisition and Agreements Management, Acquisition Management Division (AMD).

2. Nature and/or description of the action being approved:

Noncompetitive purchase of a non-commercial product. NIST intends to issue a Sole Source Award to Auburn University, 321-A Ingram Hall, Auburn, Alabama 36849, under the authority of 41 U.S.C. 3304(a)(1).

3. A description of the supplies or services required to meet the agency's needs (including estimated value):

The NIST Material Measurement Laboratory, Applied Chemicals and Materials Division, Fatigue and Fracture Group (FFG) in Boulder, Colorado has a requirement for free-standing micro specimens of electroplated copper and micro sized lead-free solder for mechanical testing to fracture. These custom micro specimens will provide NIST with appropriate specimens of NIST's design, for NIST to perform tests to fracture to measure various mechanical properties of the materials.

This will help FFG meet its mission of providing critical material properties, which are currently lacking or sparse, to the U.S. semiconductor advanced packaging industry to improve the accuracy of their finite element models and physics-based models; this will support the industry's R&D efforts and ensure reliability of its products.

For the U.S. semiconductor industry to improve its chip interconnect performance and manufacturing processes and to lower its costs, it needs to use more accurate material data. The goal of this FFG CHIPS Metrology project is to provide the industry with that data. To do this, FFG will perform measurements on material specimens that mimic the interconnects as closely as possible in terms of size, chemistry, microstructure and grain size. The specimens also need to be of precise geometry (designed by NIST) to ensure that the material's mechanical behavior is well-understood. Finally, the specimens also need to be attached to appropriate mechanical structures to interface them to NIST testing instruments, so the specimens also require ancillary parts such as grip sections, pin holes and handles. Meeting all these requirements requires a substantial amount of time and manpower for R&D; hence, NIST is outsourcing the specimen fabrication for this requirement.

The estimated value of this requirement is [REDACTED], to include fabrication and delivery of multiple sets of custom test specimens using NIST's designs, as well as progress reports, drafts of journal publications, and conference presentations. The period of performance will consist of a base period of 2 years plus a 1-year option period.

4. An identification of the statutory authority permitting other than full and open competition:

41 U.S.C. 3304(a)(1); this statute is being implemented for non-commercial products or services in accordance with FAR 6.103-1(b).

5. A demonstration that the proposed contractor's unique qualifications or the nature of the acquisition requires the use of the authority cited:

In accordance with FAR 6.103-1, Auburn University has been identified as the only source that produces a product and service that meets all of NIST's minimum requirements for fabrication of micro specimens of electroplated copper and micro sized lead-free solder for mechanical testing to fracture. Auburn University is the only source known to have the capability to fabricate appropriate micro-sized mechanical test specimens with geometries designed by NIST, and with chemical compositions and microstructures tailored to be similar to those of Cu hybrid bonded and solder bonded interconnects used in the heterogeneous integration industry. NIST also needs the contractor to design and incorporate other ancillary functional parts into the specimens to enable integration with a variety of NIST's measurement instruments ranging from table-top instruments to Micro Electro Mechanical Systems (MEMS). While commercial sources exist, market research indicates that they do not have the capability to fabricate free-standing micro solder specimens and free-standing electroplated copper specimens. Based on the findings from market research, Auburn University is the only entity that can meet NIST requirements.

6. A description of efforts made to ensure that offers are solicited from as many potential sources as is practicable, including whether a notice was or will be publicized as required by Subpart 5.2 and, if not, which exception under 5.202 applies:

The Contract Specialist posted Sources Sought Notice #NIST-SS25-21 from February 21, 2025, to March 14, 2025, on SAM.gov. The notice indicated that NIST was seeking to identify sources that are capable of providing micro specimens of electroplated copper and micro sized lead-free solder for mechanical testing to fracture, and invited all responsible sources interested to identify their interest and capability. By the closing date, the Contract Specialist received one response from Auburn University. Additionally, the Contract Specialist posted a combined Sources Sought Notice and Notice of Intent to Sole Source #NIST-SS26-CHIPS-04 with Auburn University on SAM.gov from November 18, 2025, to December 3, 2025, and received no responses.

7. A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:

The Contracting Officer will only award a purchase order if they determine the price to be fair and reasonable. They will rely on the Independent Government Cost Estimate, reasonably available pricing information, and their professional judgment in making that determination. Therefore, the contracting officer determines that the anticipated cost to the Government will be fair and reasonable.

8. A description of the market research conducted (see part 10) and the results or a statement of the reason market research was not conducted:

This is a specialized requirement for custom products, resources, and expertise. Market research was conducted from November 2024 to December 2025 to identify available sources and to assess the availability of the marketplace to meet the requirement as follows:

- a. Conducted the following internet searches: general internet search, GSA, and Thomas Register. None of the companies in the search results provide microfabrication services from all the materials and the design NIST requires.

- b. Attended two conferences, IEEE PAINE in November 2024 and IMAPS Symposium in September 2025, and spoke informally with several companies and universities about their fabrication capabilities, but none can meet NIST's specific requirements of providing custom micro specimens of electroplated copper and micro sized lead-free solder for mechanical testing to fracture as described in Section 3.
- c. Posted a Sources Sought notice (NIST-SS25-21) on SAM.gov from 2/21/25 to 3/14/25. The only response received was from Auburn University.
- d. Posted a second Sources Sought Notice combined with a Notice of Intent to Sole Source (NIST-SS26-CHIPS-04) on SAM.gov from 11/18/25 to 12/03/25. No responses were received.

9. **Any other facts supporting the use of other than full and open competition:**

The requiring activity has verified that the requirement, as stated in the specification, represents its minimum requirement.

10. **A listing of the sources, if any, that expressed, in writing, an interest in the acquisition:**

Only one (1) source expressed an interest: Auburn University.

11. **A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:**

The AMD will closely scrutinize any requirements for subsequent acquisitions of this type or requirement and will ensure that any restrictions to competition are adequately documented and supported before proceeding.

CERTIFICATION:

Requisitioner:

I certify that the facts and representations under my cognizance, which are included in this justification, and which form a basis for this justification, are complete and accurate.

[REDACTED] [REDACTED]
[REDACTED], Contracting Officer's Representative

Contracting Officer:

I certify that the justification is accurate and complete to the best of my knowledge and belief.

[REDACTED] [REDACTED]
[REDACTED], Contracting Officer

Approved by:

[REDACTED] [REDACTED]
[REDACTED], NIST Competition Advocate

Approved by:

[REDACTED] [REDACTED]
[REDACTED], DOC Competition Advocate